

1 — Import Libraries

```
import pandas as pd
import numpy as np

from sklearn.preprocessing import StandardScaler
from sklearn.decomposition import PCA
from sklearn.model_selection import train_test_split
from sklearn.ensemble import RandomForestClassifier
from sklearn.metrics import accuracy_score
```

2 — Load Dataset

```
data = pd.read_csv("Wine_Quality.csv")
```

3 — Explore Dataset

```
print(data.head())
```

```

    fixed acidity  volatile acidity  citric acid  residual sugar  chlorides \
0             7.4              0.70         0.00             1.9        0.076
1             7.8              0.88         0.00             2.6        0.098
2             7.8              0.76         0.04             2.3        0.092
3            11.2              0.28         0.56             1.9        0.075
4             7.4              0.70         0.00             1.9        0.076

    free sulfur dioxide  total sulfur dioxide  density    pH  sulphates \
0                11.0              34.0    0.9978  3.51        0.56
1                25.0              67.0    0.9968  3.20        0.68
2                15.0              54.0    0.9970  3.26        0.65
3                17.0              60.0    0.9980  3.16        0.58
4                11.0              34.0    0.9978  3.51        0.56

    alcohol  quality
0         9.4        5
1         9.8        5
2         9.8        5
3         9.8        6
4         9.4        5
```

```
print(data.info())
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 1599 entries, 0 to 1598
Data columns (total 12 columns):
 #   Column              Non-Null Count  Dtype
---  -
 0   fixed acidity        1599 non-null   float64
 1   volatile acidity     1599 non-null   float64
 2   citric acid          1599 non-null   float64
 3   residual sugar       1599 non-null   float64
 4   chlorides            1599 non-null   float64
 5   free sulfur dioxide  1599 non-null   float64
 6   total sulfur dioxide 1599 non-null   float64
 7   density              1599 non-null   float64
 8   pH                  1599 non-null   float64
 9   sulphates           1599 non-null   float64
10   alcohol              1599 non-null   float64
11   quality              1599 non-null   int64
dtypes: float64(11), int64(1)
memory usage: 150.0 KB
None
```

```
print(data.shape)
```

```
(1599, 12)
```

```
print(data.describe())
```

```

    fixed acidity  volatile acidity  citric acid  residual sugar \
count  1599.000000      1599.000000  1599.000000      1599.000000
mean     8.319637         0.527821    0.270976         2.538806
std     1.741096         0.179060    0.194801         1.409928
min     4.600000         0.120000    0.000000         0.900000
25%     7.100000         0.390000    0.090000         1.900000
50%     7.900000         0.520000    0.260000         2.200000
75%     9.200000         0.640000    0.420000         2.600000
max    15.900000        1.580000    1.000000        15.500000
```

	chlorides	free sulfur dioxide	total sulfur dioxide	density \
count	1599.000000	1599.000000	1599.000000	1599.000000
mean	0.087467	15.874922	46.467792	0.996747
std	0.047065	10.460157	32.895324	0.001887
min	0.012000	1.000000	6.000000	0.990070
25%	0.070000	7.000000	22.000000	0.995600
50%	0.079000	14.000000	38.000000	0.996750
75%	0.090000	21.000000	62.000000	0.997835
max	0.611000	72.000000	289.000000	1.003690

	pH	sulphates	alcohol	quality
count	1599.000000	1599.000000	1599.000000	1599.000000
mean	3.311113	0.658149	10.422983	5.636023
std	0.154386	0.169507	1.065668	0.807569
min	2.740000	0.330000	8.400000	3.000000
25%	3.210000	0.550000	9.500000	5.000000
50%	3.310000	0.620000	10.200000	6.000000
75%	3.400000	0.730000	11.100000	6.000000
max	4.010000	2.000000	14.900000	8.000000

4 — Define Features and Target

```
X = data.drop("quality", axis=1)
y = data["quality"]
```

5 — Feature Scaling

```
scaler = StandardScaler()

X_scaled = scaler.fit_transform(X)
```

6 — Apply PCA

```
pca = PCA(n_components=10)

X_pca = pca.fit_transform(X_scaled)
```

7 - Train Test Split

```
X_train, X_test, y_train, y_test = train_test_split(
    X_pca, y, test_size=0.2, random_state=42
)
```

8 - Train Model

```
model = RandomForestClassifier(n_estimators=300)

model.fit(X_train, y_train)
```

```
▼ RandomForestClassifier ⓘ ?
RandomForestClassifier(n_estimators=300)
```

9 - Prediction

```
y_pred = model.predict(X_test)
```

10 — Accuracy

```
accuracy = accuracy_score(y_test, y_pred)

print("Model Accuracy:", accuracy)
```

```
Model Accuracy: 0.690625
```

